Chamber fibers, reflections and regular covers

Human Compression project

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Abstract

Two elementary descriptions govern the results considered here. Normalizing part of a flag turns its remaining choices into a double-coset space. Normalizing the image of a group intersection turns its remaining obstruction into a kernel intersection. We prove these statements, give an explicit chamber model for a reflection interchanging two regular group orbits, and construct the smallest regular cover by intersecting a relation kernel with its mirror. The hypertope and polytope applications supply different geometric hypotheses; none of those hypotheses is inferred from a finite order calculation. The arguments rework previously reviewed proofs and retain the standard correspondence between string rotation groups and polytopes at its precise point of use.

1 Normalizing an incidence flag

An incidence geometry consists of a set of objects, a finite type set I, and a reflexive symmetric incidence relation for which distinct objects of the same type are not incident. A flag is a set of pairwise incident objects. We require every flag to extend to a chamber, that is, a flag containing one object of every type. A panel omits exactly one type. The geometry is thin if each panel is contained in exactly two chambers. The residue of a flag consists of the objects incident with it, of types not present in the flag, with the induced incidence relation. Residual connectedness means that every residue of rank at least two, including the whole geometry, has a connected incidence graph.

Let G be a group and let $H_i \le G$ for $i \in I$. We use left cosets gH_i as objects of type i, with incidence given by nonempty intersection. This construction always gives an incidence system; when it is called a geometry below, the flag-extension condition is an explicit hypothesis. The action is left multiplication. Write

$$H_J = \bigcap_{j\in J} H_j, \qquad H_\varnothing = G, \qquad S_{J,k} = \bigcap_{j\in J} H_j H_k \quad (k \notin J).$$

The product H_jH_k is a set product; it need not be a subgroup.

Theorem 1.1 (The extension fiber)**.** *Suppose G is transitive on flags of type J. Then its orbits on flags of type $J \cup \{k\}$ are naturally in bijection with*

$$H_J \backslash S_{J,k} / H_k.$$

Here the expression denotes the orbits of H_J on the set of cosets xH_k contained in $S_{J,k}$. The stabilizer of the extension represented by xH_k is $H_J \cap xH_kx^{-1}$.

Proof. Move the J-part of a flag to $\{H_j : j \in J\}$. A type-k object xH_k meets H_j exactly when $x \in H_jH_k$. Consequently the possible remaining objects are precisely $S_{J,k}/H_k$. Two normalizations of the same J-flag differ by an element of its stabilizer H_J, so the indicated orbit space is independent of the normalization. Conversely an element relating two normalized extensions fixes their common J-part and therefore lies in H_J. This proves both directions of the orbit correspondence. Fixing the last object adds exactly the condition $g \in xH_kx^{-1}$ to $g \in H_J$, giving the stabilizer formula. □

Corollary 1.2. *If $H_J = 1$, every normalized extension has trivial stabilizer, and the number of flag orbits is the number of cosets in $S_{J,k}/H_k$. In particular there are exactly two such orbits precisely when*

$$S_{J,k} = H_k \dot\cup xH_k \quad \text{for some } x \notin H_k.$$

If H_k is finite, this is equivalent to $|S_{J,k}| = 2|H_k|$.

Proof. The action of the trivial stabilizer makes each normalized extension an orbit. The identity belongs to every set product, so H_k is one of the cosets. All cosets have size $|H_k|$; when this is finite their number is determined by the displayed cardinality. No such division of infinite cardinals is used. □

The normalization argument is the usual orbit–stabilizer argument applied to an extension of a flag. It contains the two-orbit double-coset calculation in Zhang–Leemans, Theorem 3.2. We use the form with arbitrary stabilizer because it states exactly which hypothesis permits the simpler two-coset test.

2 Reconstructing a reflection from two chamber orbits

A type-preserving map of chambers preserves an i-adjacency when it preserves the operation of replacing the type-i object in a chamber. In a thin geometry this operation has a unique value different from the original chamber.

Lemma 2.1 (Propagation and reconstruction)**.** *In a residually connected incidence geometry, the chambers containing any fixed flag are connected by adjacency steps that retain that flag. If the geometry is thin, a type-preserving automorphism fixing a chamber is the identity. Moreover, every permutation of chambers preserving every adjacency color comes from a unique type-preserving automorphism of the geometry.*

Proof. We first prove chamber connectedness by induction on the rank of a residue. A residue of rank zero has one chamber. In rank one any two chambers are adjacent. At higher rank its incidence graph is connected. Join an object in the first chamber to an object in the second by an incidence path. Consecutive objects lie together in a chamber, since every flag extends. Successive chambers chosen in this way share an object; the induction hypothesis in that object's residue joins them while retaining it. Concatenating these paths joins the original chambers. Applying the same argument inside the residue of a fixed flag proves the first assertion.

In a thin geometry a chamber-fixing automorphism fixes the unique adjacent chamber of each color. Induction along a chamber path fixes every chamber and hence every object. For reconstruction, the chambers containing a type-i object form exactly one connected component after all i-edges are removed. One inclusion follows because the remaining moves preserve that object; the opposite inclusion is the first assertion. A color-preserving chamber permutation therefore permutes these components, inducing a bijection on the objects of each type. Two objects are incident exactly when they occur in a common chamber, again by flag extension. Thus the induced map preserves incidence in both directions. Its chamber action is the prescribed one, so it is unique. □

Theorem 2.2 (Two regular orbits)**.** *Let $G = \langle \alpha_i : i \in I \rangle$, where $0 \in I$ and $\alpha_0 = 1$. Suppose a thin, residually connected geometry has chambers*

$$(g, +),\ (g, -) \qquad (g \in G)$$

with G acting by left multiplication in each copy, and with its i-adjacencies given by

$$(g, +) \ \sim_i \ (g\alpha_i^{-1}, -). \tag{1}$$

Then its full type-preserving automorphism group is G unless an automorphism $\theta \in \mathrm{Aut}(G)$ satisfies

$$\theta(\alpha_i) = \alpha_i^{-1} \quad (i \in I). \tag{2}$$

If such θ exists, the full group is $G \rtimes \langle F \rangle$, where $F^2 = 1$, $FgF^{-1} = \theta(g)$, and

$$F(g,+) = (\theta(g),-), \qquad F(g,-) = (\theta(g),+).$$

In the first case the geometry is chiral; in the second it is regular.

Proof. An automorphism satisfying (2) squares to the identity on every generator, and hence on G. The displayed map F is therefore an involution. An edge in (1) maps to

$$(\theta(g),-) \ \sim_i \ (\theta(g)\alpha_i,+),$$

which is an edge by the same formula. Lemma 2.1 extends F to an automorphism of the geometry. Direct evaluation on both chamber copies gives $FgF^{-1} = \theta(g)$.

Conversely, let A be the full type-preserving automorphism group. It acts freely on chambers by Lemma 2.1. Evaluation at $(1,+)$ identifies its elements whose image lies in the positive copy exactly with G. If A also reaches the negative copy, compose one such element with a member of G to obtain F taking $(1,+)$ to $(1,-)$. This normalized element is unique. Since 0-adjacency exchanges these two chambers, F also takes $(1,-)$ to $(1,+)$; freeness gives $F^2 = 1$. There are exactly two cosets of G in A, so G is normal.

Write $\theta(g) = FgF^{-1}$. The chamber $(\alpha_i,+)$ is obtained from $(1,+)$ by a 0-step and then an i-step. Applying F gives the i-neighbor of $(1,+)$, namely $(\alpha_i^{-1},-)$. On the other hand $F(\alpha_i,+) = \theta(\alpha_i)(1,-)$. Freeness on the negative copy yields (2). Thus orbit fusion is equivalent to the specified inversion, and without it the full group has exactly the two alternating chamber orbits. No finite cardinality assumption entered the argument. □

This is an explicit version of the marked-inversion criterion in Fernandes–Leemans–Weiss, *Highly symmetric hypertopes*, Theorem 8.2. The formula for F discharges the extension from a group automorphism to an incidence automorphism. It does not identify an arbitrary change of matrix coordinates with a geometric reflection.

3 Intersections in a cover and its mirror

The previous section studies a possible symmetry of one chamber geometry. For regular covers the same issue is expressed at the level of defining relations: the relations must be invariant under a prescribed reflection. We first separate the intersection calculation from that geometric interpretation.

Theorem 3.1 (Intersection transport)**.** *Let $\pi : H \to G$ be a homomorphism and let $C \le A \cap B$ be subgroups of H. If $\pi(A) \cap \pi(B) = \pi(C)$, then*

$$A \cap B = C \quad \Longleftrightarrow \quad A \cap B \cap \ker\pi \le C.$$

No finiteness, centrality or surjectivity hypothesis is required.

Proof. The forward implication is immediate. Given $x \in A \cap B$, the image hypothesis supplies $c \in C$ with $\pi(c) = \pi(x)$. Then $xc^{-1} \in A \cap B \cap \ker\pi$. The condition on this kernel puts xc^{-1} in C, and therefore puts x in C. The other inclusion was a hypothesis. □

Theorem 3.2 (The invariant relation kernel)**.** *Let U be a group, let $\mu \in \mathrm{Aut}(U)$ satisfy $\mu^2 = 1$, and let $N \trianglelefteq U$. Set*

$$\overline{N} = \mu(N), \quad \Gamma = U/N, \quad \overline{\Gamma} = U/\overline{N}, \quad \Lambda = \mathrm{im}\big(U \longrightarrow \Gamma \times \overline{\Gamma}\big).$$

Then $\Lambda \cong U/(N \cap \overline{N})$. The intersection $N \cap \overline{N}$ is the largest μ-invariant normal subgroup contained in N. Thus every quotient U/J with $J \le N$ and $\mu(J) = J$ maps to Λ over Γ.

The product $N\overline{N}$ is the smallest μ-invariant normal subgroup containing N. Its image $X = (N\overline{N})/N$ in Γ satisfies

$$X = \{g \in \Gamma : (g, 1) \in \Lambda\}. \tag{3}$$

In particular $X = 1$ if and only if $N = \overline{N}$.

Proof. The diagonal map has kernel $N \cap \overline{N}$, giving the first isomorphism. Applying μ exchanges the two factors of this intersection. If $J \leq N$ and $\mu(J) = J$, applying μ to its inclusion gives $J \leq \overline{N}$, proving maximality and the factorization. The two normal subgroups $N, \overline{N}$ have normal product; applying μ exchanges its factors, and normality makes their two products equal. Every invariant normal subgroup containing N also contains $\overline{N}$, hence their product.

A pair $(g, 1)$ in the diagonal image is represented by some $u \in \overline{N}$. Its first coordinate ranges exactly over the image of $\overline{N}$ in U/N, which is $(N\overline{N})/N$. Finally $X = 1$ gives $\overline{N} \leq N$. Applying μ gives the reverse inclusion. The converse is immediate. □

3.1 The geometric input for string rotation groups

We state the geometric correspondence used in the cover application rather than hide it in the word "group". Let

$$W_n = \langle r_0, \dots, r_{n-1} \mid r_i^2 = 1, [r_i, r_j] = 1 \ (|i - j| > 1)\rangle, \qquad U_n = W_n^+, \qquad s_i = r_{i-1} r_i.$$

Here W_n^+ is the subgroup of words of even length and $[x, y] = x^{-1}y^{-1}xy$. The generators of U_n have the presentation

$$U_n = \langle s_1, \dots, s_{n-1} \mid (s_i \cdots s_j)^2 = 1 \ (i < j)\rangle.$$

For a marked quotient $\Gamma = \langle \sigma_1, \dots, \sigma_{n-1}\rangle$, put

$$\Gamma_{n-1} = \langle \sigma_1, \dots, \sigma_{n-2}\rangle, \quad \Gamma_0 = \langle \sigma_2, \dots, \sigma_{n-1}\rangle, \quad \Gamma_{0,n-1} = \langle \sigma_2, \dots, \sigma_{n-2}\rangle.$$

The standard string C^+ correspondence identifies the rotation groups of chiral or directly regular abstract polytopes with these marked quotients satisfying the intersection property. In its recursive form, for rank at least three the two end groups must themselves satisfy the property and must intersect exactly in the medial group. The rank-two group is a nontrivial cyclic group, finite or infinite; the rank-one rotation group is trivial. This is the correspondence of Schulte–Weiss, with the recursive form cited by Cunningham as Definition 2.1. We use the mathematically consistent rank-two base, correcting the rank-one line printed in the checked author edition.

The mirror is induced by conjugation by r_0:

$$\mu(s_1) = s_1^{-1}, \qquad \mu(s_2) = s_1^2 s_2, \qquad \mu(s_i) = s_i \quad (i > 2). \tag{4}$$

Direct regularity is equivalent to invariance of the relation kernel under this mirror, or equivalently to the corresponding automorphism of the marked quotient. Marked epimorphisms give covers. Without the intersection property, the pertinent quotients describe oriented maniplexes rather than necessarily polytopes. These correspondence statements are retained geometric theorems, not conclusions of Theorem 3.1.

For an actual chiral polytope with group U_n/N, Theorem 3.2 constructs the smallest regular maniplex covering it: the rotation group is Λ. Indeed the intersection kernel is mirror invariant, and every regular cover has an invariant kernel contained in N. All regular polytope covers of a chiral polytope are included. To see the required orientability, color the flags of the chiral polytope by its two automorphism orbits and pull that coloring back along the covering map; every adjacency changes color. Regularity then identifies this bipartition with the even-word subgroup. There is no assumption of a finite flag set.

3.2 Local mirrors and embedded kernels

Let $A, B, C \leq \Lambda$ be generated by the paired first $n-2$, last $n-2$, and middle $n-3$ rotation generators. Let K, L, M denote the facet, vertex-figure and medial section of the original polytope, with compatible base flags. In the first factor Γ, the three section groups are actual subgroups, not just isomorphism types. Write $X(K), X(L), X(M)$ for their chirality groups.

Lemma 3.3 (Local mirror compatibility)**.** *The groups A and B are the mirror mixes of the compatibly marked end sections, up to conjugating their second coordinates. For the second projection $\pi : \Lambda \to \overline{\Gamma}$, their kernels and the medial kernel are*

$$A \cap \ker \pi = X(K) \times 1, \quad B \cap \ker \pi = X(L) \times 1, \quad C \cap \ker \pi = X(M) \times 1. \tag{5}$$

Proof. Identify $\overline{\Gamma}$ with Γ using the isomorphism induced by μ. In these coordinates the paired generators are $(\sigma_i, \mu(s_i)N)$. On the universal facet subgroup, conjugation by r_0 is its own intrinsic mirror. Formula (3) immediately gives the assertion for A.

The intrinsic mirror on the universal vertex subgroup $\langle s_2, \dots, s_{n-1} \rangle$ is conjugation by r_1. Since $r_0 = s_1 r_1$, conjugation by r_0 is conjugation by r_1 followed by conjugation by s_1. Conjugating the second coordinate by σ_1^{-1} therefore turns the global mirror marking into the intrinsic local mirror marking. This changes neither the first coordinate nor whether the second coordinate is the identity. The same calculation applies to the medial universal subgroup $\langle s_2, \dots, s_{n-2} \rangle$. Applying (3) to these two local marked groups proves the remaining identities. This argument does not assert that a chiral section has a mirror automorphism of its own group. □

Lemma 3.4 (Prime degree and minimality)**.** *Suppose R is the smallest regular maniplex covering a polytope P, and R is not a polytope. If a finite regular polytope Q covers R with prime degree, then Q is a minimal regular polytope cover of P. If R is itself a polytope, it is the unique minimal regular polytope cover of P, without a finiteness assumption.*

Proof. Every regular polytope cover of P factors through R by smallestness. An intermediate regular cover under Q therefore corresponds to an intermediate normal subgroup of the prime-order kernel of $\Gamma(Q) \to \Gamma(R)$ (or of the rotation groups in the orientable case). It is either trivial or the whole kernel. The second possibility gives R, which is not polytopal, so the only polytope under Q in this order is Q itself. If R is polytopal, it lies under every regular polytope cover of P and is consequently the unique minimal one. This latter argument uses no group orders. □

4 Attribution and proof boundaries

The double-coset argument is a general normalization of a group action. The marked-inversion theorem makes explicit the chamber permutation behind the criterion in Fernandes–Leemans–Weiss (2016). The invariant kernel and mirror mix are standard constructions in the theory of chiral covers; Cunningham's 2023 author manuscript uses them. The intersection-transport proof has also been used in the earlier proof bank. Their value here is that the applications can invoke these fully stated interfaces without repeating the orbit and kernel arguments. No claim of historical priority is attached to this organization.

The retained geometric inputs are the string C^+ correspondence, its recursive intersection criterion, the marked-cover interpretation and the Schulte–Weiss theorem that facets of facets and vertex-figures of vertex-figures of a chiral polytope are directly regular. The last theorem is used in the application, not in the elementary kernel proof. The standard affine description of square torus maps and the specified older classification of atomic tight polytopes are additional application inputs, stated there.

Formal certificates cover only their declared algebraic and action statements. The seven new declarations in `TwoOrbitReflection.lean` check the actual two-copy formulas, including involutivity from the generating and inversion hypotheses. Two unchanged modules provide seven further

compiled declarations; their declaration-to-argument map distinguishes the used intersection and propagation interfaces from unused statements in those files. They do not formalize the whole incidence-geometry definition, the string C^+ correspondence, the older classification or the six finite groups. The complete required GAP program for those groups is supplied separately. Its trusted exact group algorithms form a computational premise for that one finite example; the general theorems and the infinite-family deductions are written proofs. Later results on the existence or multiplicity of regular covers are separate results, not premises silently imported from the three earlier cover manuscripts.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Mathematicae 90 (2016), 1045–1067. Theorem 8.2.

[2] W.-J. Zhang and D. Leemans, *Testing chirality on hypertopes*, arXiv:2511.14562v1 (November 2025); subsequently Journal of Algebraic Combinatorics 63 (2026), article 61.

[3] G. Cunningham, *Chiral polytopes whose smallest regular cover is a polytope*, author manuscript, 25 July 2023; Journal of Combinatorial Theory, Series A 204 (2024), 105839. The checked edition is the eight-page author manuscript.

[4] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS Series 4 (1991), 493–516. The correspondence and section theorem are retained as cited by the source.

[5] G. Cunningham and D. Pellicer, *Tight chiral polytopes*, Journal of Algebraic Combinatorics 54 (2021), 837–878. The specified Table 4 families and published Proposition 11 are retained; the acquired arXiv excerpt labels the latter Proposition 4.2.